



RAM-Stack: A ROS 2 Autonomy Stack for Multi-Agent SLAM and Navigation

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I. Overview

This research presents the development of RAM-Stack (ROS Autonomous Multi-Agent Stack), a ROS 2-based autonomy stack for multi-agent systems incorporating components such as Navigation2 (Nav2), SLAM Toolbox, and ROS 2 Control. This system facilitates robust autonomous navigation in both hardware and simulation environments. This integrated approach not only enhances the performance and reliability of autonomous systems but also serves as a versatile framework for future research and development in robotic autonomy.

II. Components

Figure 1 shows the structure of the autonomy stack with the following components:

Motion Commands — Generates desired motion commands as a 3D twist (i.e., linear and angular velocities). The Twist Mux forwards the highest-priority command to the hardware system.

Feedback System — Provides necessary state feedback on velocity and position for closed-loop speed control and mapping.

Hardware System — Translates and executes the desired velocity into motion commands that can be interpreted by the physical hardware, such as applied voltages.

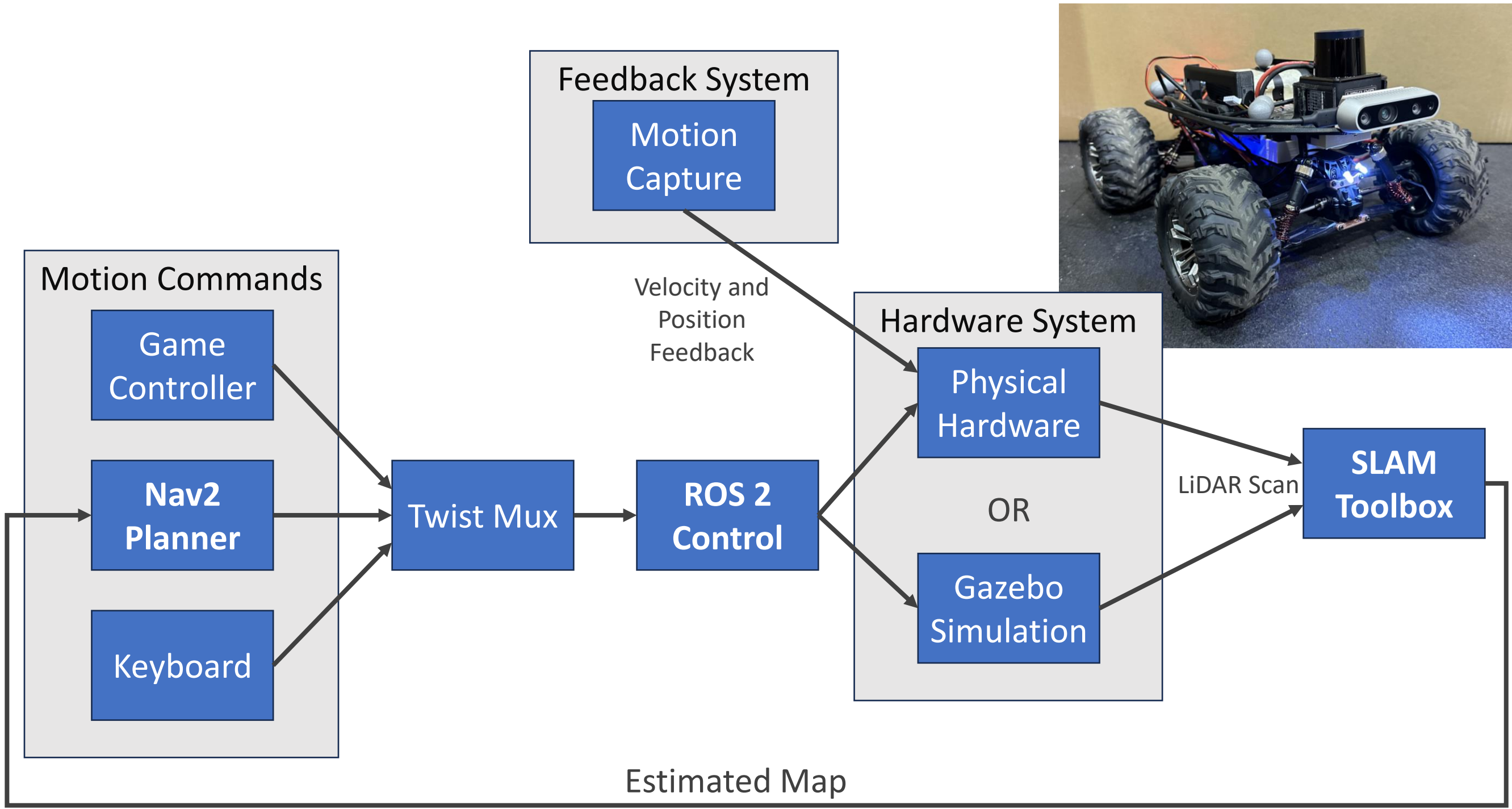


Figure 1: The process diagram for the ROS 2-based autonomy stack

III. ROS 2 Control, SLAM Toolbox, and Nav2

ROS 2 Control converts rigid-body twist commands into wheel velocities based on the vehicle’s kinematic model.

SLAM Toolbox creates a map of the previously unknown environment while traversing the area.

Nav2 utilizes the estimated map to generate and command feasible path following while executing dynamic obstacle avoidance.

IV. Results and Discussion

Figure 2 illustrates our ground vehicle autonomously following a self-generated path (represented by the green line) using a map created through SLAM in the AutonomyNM high bay. This map effectively identifies multiple obstacles, including boxes and barrels, throughout the environment.

This autonomy stack serves as a robust plug-and-play resource for future research in autonomy, particularly involving diverse multi-agent systems. For example, the SLAM Toolbox can be substituted with a swarm-compliant algorithm. Different vehicles, such as drones, can be seamlessly integrated by adjusting only the ROS 2 Control component. By developing this flexible framework, we aim to facilitate advancements in autonomous systems and pave the way for innovative applications in complex environments.

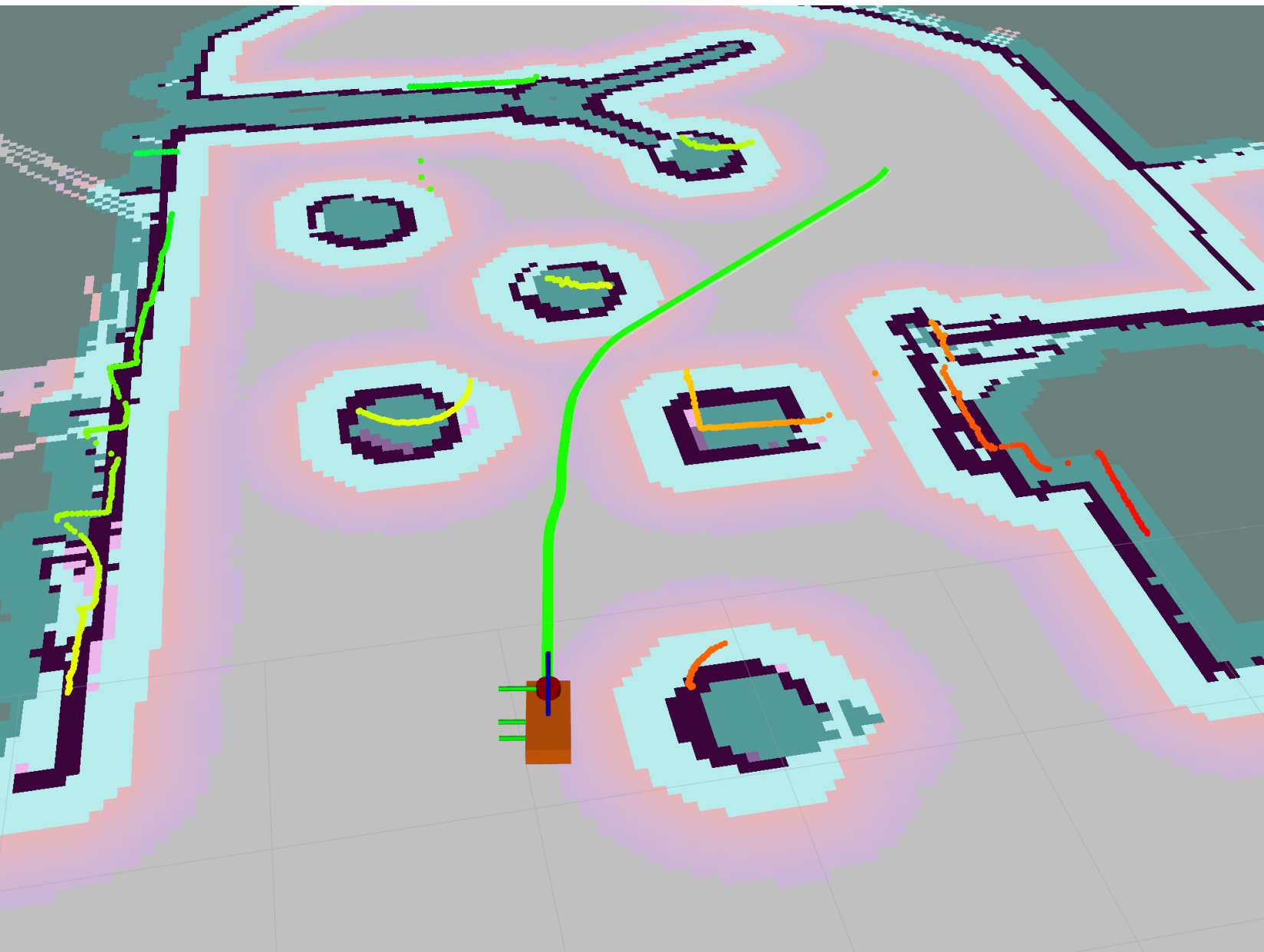


Figure 2: Visualization of our robot (orange) traversing a previously mapped environment along an optimal path (green)